

Reproduction of Fig. 3 Context-Dependent GRN Construction for Hormone-Dependent Neoplasms

Bokang Shi

Visiting Student, Imperial College London

Cell Systems Laboratory, Institute for Protein Research, Osaka University

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Supervisor: Prof. Mariko Okada

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Introduction

Gene regulatory networks (GRNs) are graphs where genes are represented as nodes and gene-gene relationships are represented as edges. In this paper, the authors compare two related ideas:

- A **context-independent GRN**, which is a general human gene-gene interaction network. It is not specific to any disease.
- A **context-dependent GRN**, which starts from the same general interaction universe but filters and weights interactions according to how relevant the supporting PubMed papers are to a specific disease context.

For this analysis, the disease context was **Neoplasms, Hormone-Dependent**, represented by the MeSH disease identifier D009376.

This disease was selected because the broad MeSH parent term Neoplasms (D009369) was not present as a disease column in the released Fig. 2/Fig. 3 Zenodo data. Neoplasms, Hormone-Dependent is an available neoplasm-related MeSH disease term in chunk 1, so it allows a single-disease Fig. 3-style reproduction without changing the underlying GRN construction method.

The aim was to reproduce the Fig. 3-style GRN construction and comparison for hormone-dependent neoplasms: generate a disease-specific context-dependent GRN, compare it against the universal context-independent GRN, and identify central genes using PageRank.

Summary of what was rerun

The raw Fig. 2 PubMed processing was **not** rerun. In the original workflow, PubMed titles and abstracts are embedded and compared with disease queries using PubMedBERT. That requires the authors' PubMed vector database environment, so this analysis used the released precomputed PubMed-disease similarity scores from Zenodo.

The Fig. 3 GRN construction step **was rerun for Neoplasms, Hormone-Dependent** using an adapted version of the original GitHub script:

Original script:

```
context-dependent-GRN/figures/fig3-0_create_context_dependent_GRN.py
```

Adapted script:

```
context-dependent-GRN/figures/fig3-0_hormone_dependent_neoplasms_create_context_dependent_GRN.py
```

The original script is designed to generate context-dependent GRNs for all MeSH diseases. The adapted script keeps the same core Fig. 3 GRN construction logic, but narrows the run to a single disease: Neoplasms, Hormone-Dependent (D009376) in chunk 1.

Data used

The analysis used the authors' released data from Zenodo:

<https://zenodo.org/records/16416117>

The raw BioREX gene interaction file was:

context-dependent-GRN/data/fig2-3/all_human_gene_interactions_2024-12-18.csv

This file contains literature-derived human gene-gene interactions. The **context-independent GRN** was generated from all unique gene pairs in this file. This network is disease-agnostic and would be the same regardless of whether the disease being studied was hormone-dependent neoplasms, plasmacytoma, breast cancer, skin cancer, or another disease.

The precomputed PubMedBERT disease-literature similarity file was:

context-dependent-GRN/data/fig2-3/All_MeSH_diseases_pmid_bert_corrs_chunk1.parquet

This file contains PubMed article relevance scores for diseases in chunk 1. Neoplasms, Hormone-Dependent is stored in this file as the column D009376.

The local MeSH annotation table identifies the disease as:

- disease label: Neoplasms, Hormone-Dependent
- MeSH ID: D009376
- chunk number: 1

How the GRNs were generated

Context-independent GRN

The context-independent GRN was generated from:

all_human_gene_interactions_2024-12-18.csv

The adapted script took all unique gene-gene pairs and treated each pair as an edge. Each edge was assigned a uniform weight of 1.0.

This means the context-independent GRN is a universal background network. It is **not disease-specific** and does not change between diseases.

Output file:

context-dependent-GRN/data/fig2-3/hormone_dependent_neoplasms_fig3_adapted/context_independent_edges.csv

Hormone-dependent neoplasms context-dependent GRN

The hormone-dependent neoplasms context-dependent GRN was generated from two inputs:

1. all_human_gene_interactions_2024-12-18.csv
2. All_MeSH_diseases_pmid_bert_corrs_chunk1.parquet

The adapted script used the D009376 column from the PubMedBERT score file. For each gene-gene interaction, it looked up the PMID supporting that interaction, mapped the PMID to its hormone-dependent neoplasm relevance score, and retained interactions with relevance score greater than 0.2.

The retained interactions were then processed using the original Fig. 3 logic:

- remove duplicate PMID-gene-pair entries
- group interactions by gene pair
- sum disease relevance scores per gene pair
- normalize weights to counts per million
- apply log1p transformation

Output files:

context-dependent-GRN/data/fig2-3/hormone_dependent_neoplasms_fig3_adapted/hormone_dependent_neoplasms_context_dependent_edges.csv

context-dependent-GRN/data/fig2-3/hormone_dependent_neoplasms_fig3_adapted/hormone_dependent_neoplasms_log1p_CPM_D009376.parquet

What the original Fig. 3 script would have done

If run unchanged, fig3-0_create_context_dependent_GRN.py would process all MeSH disease chunks:

- chunk 1
- chunk 2
- chunk 3

For every disease column in those chunks, it would generate disease-specific GRN weights. The outputs would be large all-disease matrices:

- All_MeSH_diseases_log1p_CPM_chunk1.parquet
- All_MeSH_diseases_log1p_CPM_chunk2.parquet
- All_MeSH_diseases_log1p_CPM_chunk3.parquet

Each disease would be represented as one column in those matrices. For example:

- D009376 = Neoplasms, Hormone-Dependent GRN
- D010954 = plasmacytoma GRN
- other MeSH IDs = other disease-specific GRNs

Therefore, the main purpose of the adaptation was to avoid regenerating every disease and instead run the same Fig. 3 construction logic only for Neoplasms, Hormone-Dependent.

Code adaptations and justification

The adapted script was:

context-dependent-GRN/figures/fig3-0_hormone_dependent_neoplasms_create_context_dependent_GRN.py

The adaptations were:

1. Disease restricted to Neoplasms, Hormone-Dependent

What changed: added `DISEASE_ID = "D009376"` and `DISEASE_LABEL = "Neoplasms, Hormone-Dependent"`.

Justification: the broad parent Neoplasms (D009369) was not present in the released disease score columns, while D009376 was available as a neoplasm-related MeSH disease term.

2. Chunk restricted to chunk 1

What changed: added `CHUNK = 1`.

Justification: the MeSH annotation table and the parquet columns show D009376 is stored in chunk 1.

The original threshold (0.2) and core weighting logic were kept unchanged from fig3-0: PMID score mapping, thresholding, duplicate removal, grouping by gene pair, CPM normalization, and log1p transformation.

3. Context-independent edge list written out

What changed: saved `context_independent_edges.csv`.

Justification: the original script uses the universal pair list internally. Saving it made the baseline GRN explicit for comparison.

4. Disease-specific edge list written out

What changed: saved `hormone_dependent_neoplasms_context_dependent_edges.csv`.

Justification: this made the disease-specific GRN directly inspectable.

5. PageRank added for both networks

What changed: used `networkx.pagerank(..., weight="weight")`.

Justification: PageRank is used in the Fig. 3 notebooks for representative disease networks; this adaptation applies the same centrality idea to Neoplasms, Hormone-Dependent.

6. Overlap table added

What changed: saved `top_pagerank_overlap.csv`.

Justification: this gives a clear comparison between top genes in the two networks.

The original GitHub scripts were not overwritten. The adapted script was added separately so the original repository files remained intact.

Command run

The adapted Fig. 3 script was run from the extracted GitHub repository's figures directory:

```
cd C:\Users\Bokan\Documents\context-dependent-GRNs\context-dependent-GRN\figures
..\..\venv\Scripts\python.exe
fig3-0_hormone_dependent_neoplasms_create_context_dependent_GRN.py
```

The script generated outputs in:

context-dependent-GRN\data\fig2-3\hormone_dependent_neoplasms_fig3_adapted\

PageRank calculation

PageRank was calculated using NetworkX:

```
nx.pagerank(graph, weight="weight")
```

For the context-independent GRN, all edge weights were 1.0, so PageRank reflects centrality in the general background gene-gene interaction network.

For the hormone-dependent neoplasms context-dependent GRN, edge weights were disease-specific literature-derived weights, so PageRank reflects centrality in the hormone-dependent neoplasm-weighted network.

The PageRank outputs were:

- context_independent_top_pagerank.csv
- hormone_dependent_neoplasms_context_dependent_top_pagerank.csv

Results

The adapted Fig. 3 script loaded 3,017,208 BioREX interaction rows and identified 891,603 unique gene pairs. After applying the D009376 relevance threshold, 566,209 interaction rows were retained before aggregation by gene pair.

Network statistics

Network	Nodes	Edges	Density	Largest component nodes	Largest component edges
Context-independent GRN	20,824	891,603	0.004112	20,789	891,585
Hormone-dependent neoplasms context-dependent GRN	15,118	242,472	0.002122	15,051	242,438

The hormone-dependent neoplasms context-dependent GRN was smaller and less dense than the context-independent GRN. This is expected because disease-context weighting filters the general interaction network to interactions supported by hormone-dependent neoplasm-relevant literature.

Top PageRank genes

The top 10 PageRank genes in the hormone-dependent neoplasms context-dependent GRN were:

Rank	Gene	PageRank
1	AKT1	0.009893
2	TP53	0.008191
3	CTNNB1	0.006220
4	MAPK1	0.006064
5	TGFB1	0.005761
6	MYC	0.005725

Rank	Gene	PageRank
7	NFKB1	0.005339
8	EGFR	0.004579
9	ESR1	0.004488
10	VEGFA	0.004188

The top 10 PageRank genes in the context-independent GRN were:

Rank	Gene	PageRank
1	AKT1	0.004450
2	TP53	0.003384
3	NFKB1	0.003179
4	MAPK1	0.003138
5	TNF	0.003095
6	CTNNB1	0.002857
7	TGFB1	0.002856
8	MYC	0.002397
9	IL6	0.002376
10	MTOR	0.002182

Top gene overlap

Among the top 50 PageRank genes, 42 genes were shared between the context-independent and hormone-dependent neoplasms context-dependent networks.

Genes appearing in the hormone-dependent neoplasms top 50 but not the context-independent top 50 were:

BRCA1, EREG, EZH2, IGF1, KRAS, PIK3R1, PRL, SNAI1

Genes appearing in the context-independent top 50 but not the hormone-dependent neoplasms top 50 were:

APP, CXCL8, FN1, IL10, NFE2L2, PPARG, RELA, TLR4

Interpretation

The context-independent GRN is a universal background network. It is identical across disease analyses because it is generated from all unique human gene-gene interactions without disease-specific weighting.

The hormone-dependent neoplasms context-dependent GRN is disease-specific. It starts from the same interaction universe, but interactions are filtered and weighted using disease-specific PubMedBERT relevance scores from the D009376 column.

Several highly ranked genes were shared between the two networks, including AKT1, TP53, MAPK1, NFKB1, MYC, CTNNB1, and TGFB1. These are broad signaling and cancer-associated genes, so it is expected that they remain central in both the general and disease-specific networks.

The disease-specific network also prioritized genes such as ESR1, BRCA1, IGF1, PRL, PIK3R1, and SNAI1 within its top 50 genes. This is biologically plausible for hormone-dependent neoplasms because these genes are linked to hormone signaling, cancer growth, or cancer-associated regulatory pathways.

Overall, this analysis reproduced the Fig. 3-style GRN construction and comparison for Neoplasms, Hormone-Dependent by minimally adapting the original GitHub Fig. 3 workflow to a single available neoplasm-related disease context.

Appendix: exact adapted analysis script

The exact final analysis script added for this project was:

context-dependent-GRN/figures/fig3-0_hormone_dependent_neoplasms_create_context_dependent_GRN.py

```
"""
Hormone-dependent neoplasm-specific adaptation of fig3-0_create_context_dependent_GRN.py.

The original script loops over all MeSH disease columns in all three chunks.
This adaptation keeps the same weighting logic but runs only the MeSH disease
ID D009376, "Neoplasms, Hormone-Dependent", which is stored in chunk 1.
"""

import time
from pathlib import Path

import networkx as nx
import numpy as np
import pandas as pd

DISEASE_ID = "D009376"
DISEASE_NAME = "hormone_dependent_neoplasms"
DISEASE_LABEL = "Neoplasms, Hormone-Dependent"
CHUNK = 1
THRESHOLD = 0.2
TOP_N = 50

def pair_from_genes(row: pd.Series) -> str:
    return "--".join(sorted([str(row["from_gene"]), str(row["to_gene"])])))

def graph_from_edges(edges: pd.DataFrame) -> nx.Graph:
    graph = nx.Graph()
    graph.add_weighted_edges_from(
        (row.from_gene, row.to_gene, float(row.weight))
        for row in edges.itertuples(index=False)
    )
    return graph

def graph_stats(graph: nx.Graph, label: str) -> dict:
    if graph.number_of_nodes() == 0:
        largest_nodes = 0
        largest_edges = 0
    else:
        largest = graph.subgraph(max(nx.connected_components(graph), key=len))
        largest_nodes = largest.number_of_nodes()
        largest_edges = largest.number_of_edges()

    return {
        "network": label,
        "nodes": graph.number_of_nodes(),
        "edges": graph.number_of_edges(),
        "density": nx.density(graph),
        "largest_component_nodes": largest_nodes,
        "largest_component_edges": largest_edges,
    }

def pagerank_table(graph: nx.Graph, top_n: int) -> pd.DataFrame:
    pagerank = nx.pagerank(graph, weight="weight")
    return (
```

```

pd.DataFrame(pagerank.items(), columns=["gene", "pagerank"])
.sort_values("pagerank", ascending=False)
.head(top_n)
.reset_index(drop=True)
)

def main() -> None:
start = time.time()
project_root = Path(__file__).resolve().parents[1]
data_dir = project_root / "data" / "fig2-3"
output_dir = data_dir / f"{DISEASE_NAME}_fig3_adapted"
output_dir.mkdir(parents=True, exist_ok=True)

relations_path = data_dir / "all_human_gene_interactions_2024-12-18.csv"
pmid_corrs_path = data_dir / f"All_MeSH_diseases_pmid_bert_corrs_chunk{CHUNK}.parquet"

print(f"Loading BioREX gene interactions: {relations_path}")
relations_df = pd.read_csv(relations_path)
relations_df["pair"] = relations_df.apply(pair_from_genes, axis=1)
all_pairs = np.array(sorted(relations_df["pair"].unique()))
print("Raw interaction rows:", len(relations_df))
print("Unique gene pairs:", len(all_pairs))

print(f"Loading Fig. 2 PubMedBERT scores: {pmid_corrs_path}")
pmid_corrs = pd.read_parquet(pmid_corrs_path, columns=["pmid", DISEASE_ID])
pmid_to_corr = pmid_corrs.set_index("pmid")[DISEASE_ID]

disease_relations_df = relations_df.copy()
disease_relations_df["corr"] = disease_relations_df["pmid"].map(pmid_to_corr)
disease_relations_df =
disease_relations_df.dropna(subset=["corr"]).reset_index(drop=True)
disease_relations_df = disease_relations_df[disease_relations_df["corr"] >
THRESHOLD].reset_index(drop=True)
print(f"[{DISEASE_ID}] Interactions after thresholding:", len(disease_relations_df))

disease_relations_df = disease_relations_df.drop_duplicates(subset=["pmid",
"pair"]).reset_index(drop=True)
disease_edges = (
disease_relations_df[["pair", "corr"]]
.groupby("pair")
.agg({"corr": "sum"})
.reset_index()
)
disease_edges["corr"] = disease_edges["corr"] / disease_edges["corr"].sum() * 1e6
disease_edges["corr"] = np.log1p(disease_edges["corr"].values)
disease_edges.columns = ["pair", "weight"]
disease_edges[["from_gene", "to_gene"]] = disease_edges["pair"].str.split("--",
expand=True)
disease_edges = disease_edges[["from_gene", "to_gene", "weight", "pair"]]

pair_to_weight = dict(zip(disease_edges["pair"], disease_edges["weight"]))
matrix_df = pd.DataFrame(
{
"pair": all_pairs,
DISEASE_ID: np.frompyfunc(lambda x: pair_to_weight.get(x, 0), 1,
1)(all_pairs).astype(np.float32),
}
)

context_independent_edges = pd.DataFrame({"pair": all_pairs})
context_independent_edges[["from_gene", "to_gene"]] =
context_independent_edges["pair"].str.split("--", expand=True)
context_independent_edges["weight"] = 1.0
context_independent_edges = context_independent_edges[["from_gene", "to_gene", "weight",
"pair"]]

print("Building graphs and calculating PageRank")
independent_graph = graph_from_edges(context_independent_edges)
dependent_graph = graph_from_edges(disease_edges)

stats = pd.DataFrame(

```



```

[
graph_stats(independent_graph, "context_independent"),
graph_stats(dependent_graph, f"{DISEASE_NAME}_context_dependent"),
]
)
independent_pr = pagerank_table(independent_graph, TOP_N)
dependent_pr = pagerank_table(dependent_graph, TOP_N)

independent_top = set(independent_pr["gene"])
dependent_top = set(dependent_pr["gene"])
overlap = pd.DataFrame(
{
"shared_top_genes": ["", ".join(sorted(independent_top & dependent_top))"],
f"{DISEASE_NAME}_only_top_genes": ["", ".join(sorted(dependent_top -
independent_top))"],
"context_independent_only_top_genes": ["", ".join(sorted(independent_top -
dependent_top))"],
"n_shared": [len(independent_top & dependent_top)],
"top_n": [TOP_N],
}
)

matrix_df.to_parquet(output_dir / f"{DISEASE_NAME}_log1p_CPM_{DISEASE_ID}.parquet",
index=False)
context_independent_edges.to_csv(output_dir / "context_independent_edges.csv",
index=False)
disease_edges.to_csv(output_dir / f"{DISEASE_NAME}_context_dependent_edges.csv",
index=False)
stats.to_csv(output_dir / "network_stats.csv", index=False)
independent_pr.to_csv(output_dir / "context_independent_top_pagerank.csv", index=False)
dependent_pr.to_csv(output_dir / f"{DISEASE_NAME}_context_dependent_top_pagerank.csv",
index=False)
overlap.to_csv(output_dir / "top_pagerank_overlap.csv", index=False)

elapsed = time.time() - start
print("\nNetwork stats")
print(stats.to_string(index=False))
print(f"\nWrote results to {output_dir}")
print("Elapsed time:", time.strftime("%H:%M:%S", time.gmtime(elapsed)))

if __name__ == "__main__":
main()

```